

Northern University of Business and Technology Khulna

Department of Computer Science and Engineering

Project Title

MediLink – Smart Doctor Appointment & Healthcare Management System.

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MediLink – Smart Doctor Appointment & Healthcare Management System.



Figure : Hospital Management System (MediLink) App Diagram

Description:

The use case diagram of the Medilink system illustrates the interaction between the main actors: Patient, Doctor, and Admin with the system functionalities. The system is designed to provide a smooth and efficient healthcare management platform.

Patients can register into the system, log in securely, search for doctors based on specialization, and book appointments according to available time slots. Doctors can log in to their dashboard, manage their schedules, and accept or reject appointment requests from patients. Admin has full control over the system, including managing users, doctors, and monitoring system activities.

This diagram clearly represents the overall functionality of the system and shows how each user interacts with different features of the Medilink platform.

Patient Flow



Description

The Patient Flow represents the complete journey of a patient within the MediLink system. Initially, the user visits the landing page and creates an account through registration. After successful registration, the patient logs into the system and accesses the patient dashboard. From the dashboard, the patient can search for doctors based on specialization, availability, and ratings. After selecting a doctor, the patient books an appointment and completes the payment process. The system then allows the patient to attend a video consultation with the doctor. After the consultation, the patient receives a digital prescription, which is securely stored along with other medical records in the system for future access. Finally, the patient can log out of the application.

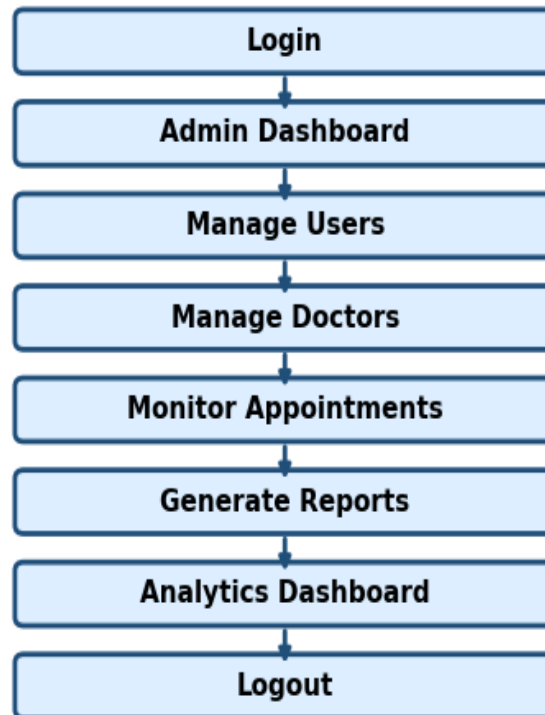
Doctor Flow



Description

The Doctor Flow describes how doctors interact with the MediLink system. Doctors first log into their accounts and access the doctor dashboard. From the dashboard, they can view upcoming appointments and access patients' information and medical history. During the consultation session, doctors can communicate with patients through video consultation and provide medical advice. After the consultation, doctors create and issue digital prescriptions. They can also update their availability and schedule to manage future appointments efficiently. Finally, doctors can securely log out from the system.

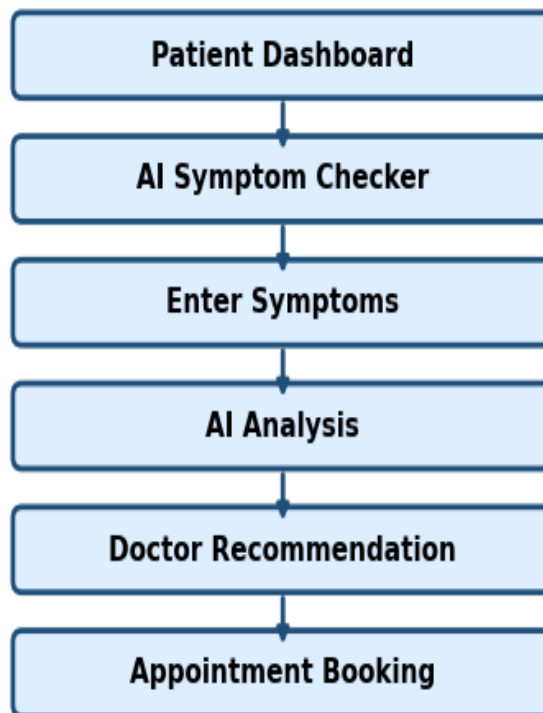
Admin Flow



Description

The Admin Flow represents the administrative functionalities of the MediLink system. The administrator logs into the system and accesses the admin dashboard. From there, the admin can manage user accounts, including patients and doctors, by adding, updating, or removing information when necessary. The administrator also monitors all appointments and system activities. Additionally, the system generates various reports and provides an analytics dashboard to help administrators monitor system performance, user activities, and healthcare statistics. After completing administrative tasks, the admin logs out from the system.

AI Symptom Checker Flow



Description

The AI Symptom Checker Flow is an intelligent feature of the MediLink system designed to assist patients before booking appointments. From the patient dashboard, users can access the AI Symptom Checker and enter their symptoms. The system analyzes the symptoms using artificial intelligence algorithms and provides preliminary health assessments. Based on the analysis, the system recommends suitable doctors or specialists for the patient's condition. The patient can then directly proceed to appointment booking with the recommended doctor, making the healthcare process faster and more convenient.

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